



ROBOTICS DREAM KEEPER CHALLENGE

# I am Korosuke!

**An era when anyone can build a robot**



uecken · my first robot · built by Robostadion (Akihabara, Tokyo)

# A fan-made karakuri robot with an RDK X5 brain

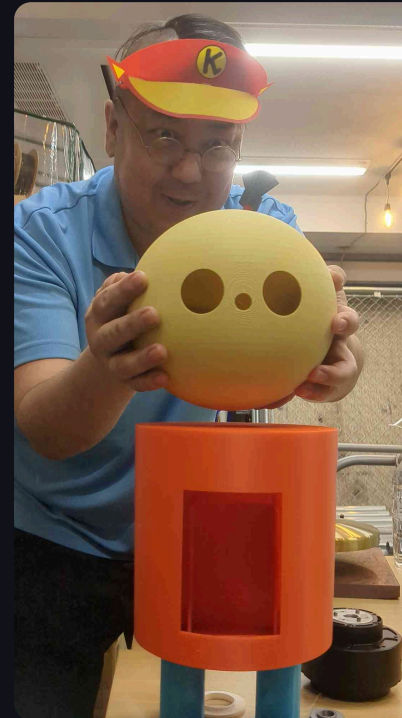
- The partner robot from *Kiteretsu Daihyakka* (Fujiko F. Fujio)
- Fan-made by **Robostadion** — brain = **one D-Robotics RDK X5**
- Mechanics / electronics / software — **designed together with AI**
- Sees · listens · thinks · talks · emotes — **100% on the board, no cloud**



## BACKGROUND

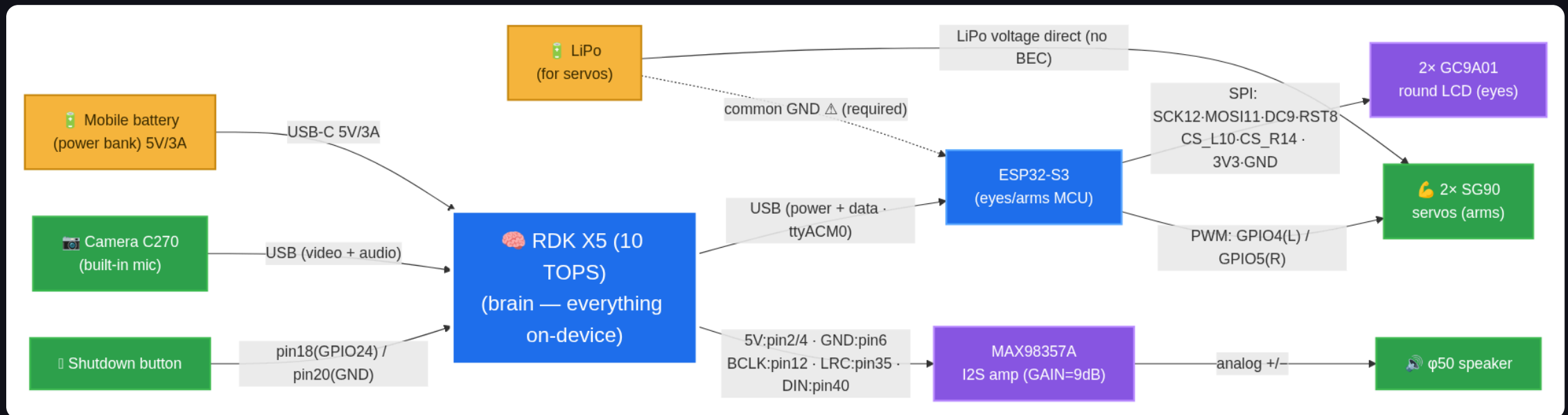
# "Why not build Korosuke?"

- I'm a **wireless engineer** — this is my first robot
- **Murata-san** (owner of Robostadion) invited me
- Saw the Discord → remembered → got the parts → **built it!**
- Inspired by **Disney's Olaf robot** & **Open Duck Mini**



▲ The "head-mounting ceremony" 🎉 — placing his head for the first time

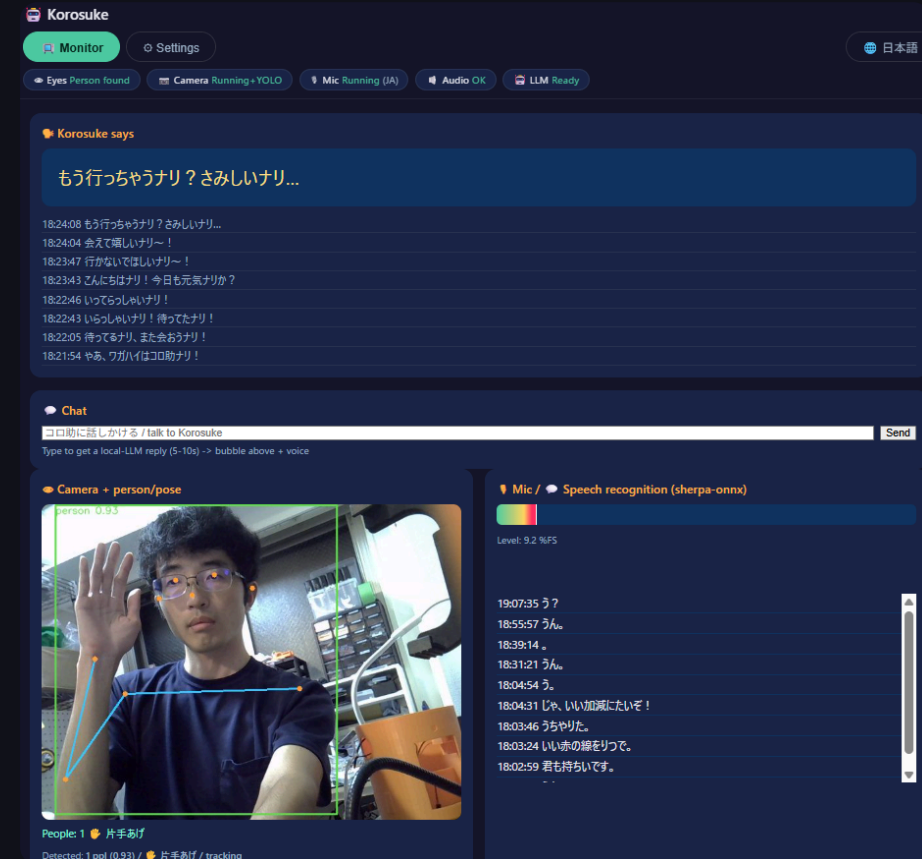
# No custom PCB — just simple wiring



**RDK X5 at the center** (all main functions) · eyes & arms via **one ESP32-S3** · servos draw **>1 A** → **separate LiPo** · tiny **I2S amp** → φ50 speaker (for size) · C270 camera = **mic** · no custom PCB

# One "brain" orchestrates everything

- 🗣️ **Listen** — speech → text (STT: SenseVoice)
- 🧠 **Think** — local LLM (TinySwallow-1.5B) · 5–10 s → "thinking eyes"
- 🗣️ **Speak** — Open JTalk → amp → speaker
- 👁️ **See** — **BPU** finds people (~20 FPS) → eyes follow you
- 💪 **Move** — ESP32-S3 → 8 eye emotions + arms



# 3D print · wire · create with AI

- 3D parts **co-created by Murata-san and AI** (v1 files in the repo)
- Head = 2 hemispheres · torso holds everything · **rope-pull arms** (rings + string)
- Wiring — exactly as the repo docs
- Software — **created together with AI** (I placed parts & debugged)
- Tools: 3D printer · soldering iron · velcro · double-sided tape · string · zip ties — **any maker space has them!**



▲ rope-pull arm — 動画をクリックで再生/停止 (click video: play/pause)



AFTERWORD

# Anyone can build a robot.

This build: **15 h printing + 15 h electronics & software  $\approx$  30 hours.**

With AI — **a custom robot in one day, maybe  $\sim$ 5 hours, is within reach!**

All open source → [github.com/gurimaruking/corosuke-robot](https://github.com/gurimaruking/corosuke-robot)



**Now — let's meet him. Korosuke, say hello!**



# 1 minute with Korosuke

1. 「コロすけ！」 → eyes find & follow
2. 「こんにちは！ 元気？」 → **thinking eyes** (local LLM, 5–10 s) → "～ナリ!" reply
3. Raise a hand → he **waves back**
4. Power button → "good night" → **XX eyes** = safe to unplug

Fallback videos: korosuke\_demo\_10s.mp4 · arm\_rope\_pull.mp4

ワガハイはコロ助ナリ！ **Thank you!** 🚶